

Supporting Information:

'Efficient Algorithms for GPU Accelerated Evaluation of the DFT Exchange-Correlation Functional'

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This supporting information contains all inputs and outputs of calculations performed using EXESS and GPU4PySCF along with the associated Python analysis scripts to generate the plots presented in the article.

1 Supporting Info Structure

```
inputs/                                # Input files and configuration
- common.py                            # Calculation parameters and molecule definitions
- exess/                               # EXESS input files for different configurations
- pyscf/                               # PySCF input files and scripts
- xyzs/                                # Molecular geometry files
  - boron_nitride/                     # BN sheet geometries
  - diamond/                           # Diamond nanoparticle geometries
  - glycine/                           # Glycine chain geometries
  - water/                             # Water cluster geometries
outputs/                               # Calculation results
- boron_nitride/                       # Results for BN systems
- diamond/                             # Results for diamond systems
- glycine/                             # Results for glycine systems
```

- water/	# Results for water systems
- exess/	# EXESS calculation outputs
- pyscf/	# PySCF calculation outputs
analysis/	# Data processing and visualization
- to_csv.py	# Convert raw outputs to CSV format
- generate_plots.py	# Superscript to generate all plots
- raw_data.csv	# Processed data summary
- plot_common.py	# Common plotting utilities
- DATA_DICTIONARY.md	# Data column descriptions for raw_data.csv
- plots/	# Generated analysis plots
requirements.txt	# Python dependencies
README.md	# Usage instructions for the python scripts